

Water Jug Problem using BFS (Python)

Simple code for AI practical examination

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from collections import deque

jug1 = int(input("Enter capacity of jug1: "))
jug2 = int(input("Enter capacity of jug2: "))
target = int(input("Enter target amount: "))

visited = set()
queue = deque()
queue.append((0,0))

while queue:
    x, y = queue.popleft()

    if (x,y) in visited:
        continue

    visited.add((x,y))
    print((x,y))

    if x == target or y == target:
        print("Target achieved")
        break

    queue.append((jug1,y))
    queue.append((x,jug2))

    queue.append((0,y))
    queue.append((x,0))

    transfer = min(x, jug2-y)
    queue.append((x-transfer, y+transfer))

    transfer = min(y, jug1-x)
    queue.append((x+transfer, y-transfer))
```